Transfer: circuits and induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply circuit relationships and explain an induction observation.

SUCCESS CRITERIA - I can...

1. Use current and voltage rules in a new circuit.
2. Account for electrical power and energy.
3. Explain induction through changing flux.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “resistance” and “power” to explain your model or result. Sentence starters are optional.

resistance: voltage divided by current for a stated operating condition

power: rate of energy transfer

induction: producing an emf through a changing magnetic flux

01 Warm-up: bring back an earlier idea

A motor is 62% efficient and the following generator is 71% efficient.
Find the combined useful fraction in sequence.

Word help: generator = a device converting mechanical input to electrical output

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $V = IR$; $P = IV$; $E = P\Delta t$; $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$.
Use stated circuit or coil conditions. Ideal battery and wires unless stated otherwise.

Worked reasoning: A 12 V supply drives a 6 ohm resistor: current 2 A and power 24 W. A complete model checks both the charge-flow rate and the energy-transfer rate.

Your turn: Draw the circuit and an energy-flow arrow. **Label** what an ammeter and voltmeter would measure and where each connects.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch two resistors in parallel across a battery and label the equal-voltage branches.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 12 V battery powers a 4 Ω resistor in series with two parallel 8 Ω resistors.
Label the network.
Calculate total current, each branch current and total input power.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A magnet is held motionless near a fixed coil, then moved into it.
Explain the change in the meter response without saying the magnet supplies unlimited energy.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Measured (I, V) pairs are (0.10, 4), (0.20, 8), (0.30, 12) in A, V.
Find resistance and explain the evidence for an ohmic model.
Draw the stated circuit and label quantities relevant to the question.

Word help: ohmic = having constant resistance under the stated physical conditions

Sentence starter: Under the assumption …, the evidence … supports … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

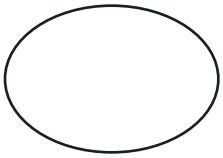
07 Exit check

A 50-turn coil has flux per turn change by 0.004 Wb in 0.20 s.
Find average emf magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 12 V source feeds a 2 ohm series resistor and a parallel pair of 4 ohm resistors.
Find total power and power in each resistor; verify the energy-rate sum.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3